

Software Requirements Specification (SRS)

for

Cloud-Based Hospital Management System (HMS)

Version 2.0

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Software Requirements Specification

Table of Contents

1	Introduction
1.1	Purpose
1.2	Scope
1.3	Definitions and Acronyms
2	Overall Description
2.1	Product Perspective
2.2	User Characteristics
2.3	Operating Environment
3	Specific Requirements
3.1	Functional Requirements (Modular)
3.2	External Interface Requirements
3.3	Non-Functional Requirements (Quality Attributes)
4	Appendix: Data Dictionary

1. Introduction

1.1 Purpose

This SRS defines the requirements for a Cloud-Based Hospital Management System (HMS). The objective is to digitize and automate all administrative, financial, and clinical operations of a multi-specialty hospital. This system will replace legacy paper-based records and siloed departmental software.

1.2 Scope

- Patient Registration & Scheduling: Online booking, queue management, and patient history.
- Electronic Health Records (EHR): Secure, centralized storage of patient medical histories, diagnoses, and treatment plans.
- Pharmacy & Inventory: Tracking of medication stock, expiry dates, and automated reordering.
- Billing & Insurance: Generation of invoices, integration with insurance providers for cashless claims.
- Staff Management: Doctor scheduling, shift management, and payroll integration.

1.3 Definitions and Acronyms

- EHR: Electronic Health Record.
- HIS: Hospital Information System.
- HL7: Health Level Seven (International standard for transferring clinical data).
- OPD: Outpatient Department.
- IPD: Inpatient Department.

2. Overall Description

2.1 Product Perspective

The HMS is a new, standalone cloud-native system (SaaS model) but must integrate with existing external legacy systems such as:

- Third-party Laboratory Information Systems (LIS).
- Government Health Registries (for vaccine tracking).
- Payment Gateways (Stripe, Razorpay).

2.2 User Characteristics

User Type	Role Description	Technical Proficiency
System Admin	Manages user roles, system configuration, and backups.	High
Doctors/Clinicians	View EHR, prescribe medication, and set treatment plans.	Medium
Nursing Staff	Update vitals, administer medication, and discharge patients.	Low to Medium
Receptionists	Handle patient check-in/out and appointment scheduling.	Low
Patients	Access a patient portal to view reports and book appointments.	Low

2.3 Operating Environment

- Platform: AWS Cloud (or Azure) with auto-scaling.
- Frontend: Web-based (React.js) and Mobile App (Flutter for iOS/Android).
- Backend: Microservices architecture (Node.js/.NET Core).

- Database: HIPAA-compliant PostgreSQL with encryption at rest.

3. Specific Requirements

3.1 Functional Requirements (Modular)

Module A: Patient Management

FR-A-01: The system must allow receptionists to register a new patient by capturing Name, DOB, Contact Info, and Emergency Contact.

FR-A-02: The system must generate a unique Patient ID (format: PAT-YYYY-XXXXX) for every registration.

FR-A-03: The system must provide a global search bar to retrieve patient records using Name, Phone, or Patient ID within < 1 second.

FR-A-04: Patients must be able to book, reschedule, or cancel OPD appointments via the Patient Portal without calling the hospital.

Module B: Electronic Health Records (EHR)

FR-B-01: Doctors must be able to document SOAP notes (Subjective, Objective, Assessment, Plan) for each visit.

FR-B-02: The system must maintain a versioned audit trail of all changes made to an EHR (who edited it, when, and what was changed).

FR-B-03: The system must allow doctors to attach diagnostic images (X-Ray, MRI) in DICOM format to the patient record.

FR-B-04: The system must flag critical allergies (e.g., Penicillin) prominently in red across all screens to prevent medication errors.

Module C: Pharmacy & Inventory Management

FR-C-01: The system must track pharmaceutical inventory in real-time based on dispensed prescriptions.

FR-C-02: The system must send an automated alert to the Procurement Manager when stock levels fall below a defined minimum threshold (reorder level).

FR-C-03: Pharmacists must be able to scan barcodes on medication packages to verify the correct drug is being dispensed against the digital prescription.

FR-C-04: The system must block the dispensing of expired drugs and quarantine them in the system.

Module D: Billing & Insurance

FR-D-01: The system must automatically generate an itemized bill (Consultation, Room Rent, Procedures, Pharmacy) upon patient discharge.

FR-D-02: The system must calculate TPA (Third-Party Administrator) insurance approvals and apply deductibles automatically.

FR-D-03: The system must support multiple payment methods: Cash, Credit/Debit Card, UPI, and Net Banking.

FR-D-04: The system must generate tax-compliant invoices (GST/IRS compliant) and send a PDF copy to the patient's registered email.

3.2 External Interface Requirements

3.2.1 Hardware Interfaces

HW-01: The system must interface with standard USB barcode scanners for pharmacy inventory checks.

HW-02: The system should integrate with biometric fingerprint scanners for secure staff attendance marking.

3.2.2 Communication Interfaces (APIs)

CI-01: The system must support HL7 FHIR (Fast Healthcare Interoperability Resources) standards to exchange data with external labs (e.g., Quest Diagnostics).

CI-02: The system must connect to an SMS Gateway (e.g., Twilio) to send appointment reminders to patients 24 hours in advance.

CI-03: The system must interface with a secure Email API to deliver medical reports securely via encrypted PDFs.

3.3 Non-Functional Requirements (Quality Attributes)

3.3.1 Legal & Regulatory Compliance (Security)

NFR-01: The system must be fully compliant with HIPAA (USA) and GDPR (Europe) regarding patient data privacy.

NFR-02: All EHR data must be encrypted using AES-256 at rest and TLS 1.3 in transit.

NFR-03: Session timeouts must occur after 10 minutes of inactivity for high-security roles (Doctors/Admins).

3.3.2 Performance & Scalability

NFR-04: The system must handle a load of up to 5,000 concurrent users during peak morning hours (9 AM - 11 AM) with a response time < 2 seconds.

NFR-05: The EHR rendering engine must display a 5-year patient history (with charts) in under 3 seconds.

3.3.3 Reliability & Disaster Recovery

NFR-06: The system must guarantee 99.99% availability for critical life-saving modules (EHR and Prescriptions) during hospital operating hours.

NFR-07: Automated database snapshots must occur every 6 hours, with a disaster recovery (DR) failover to a secondary region within 30 minutes.

3.3.4 Usability

NFR-08: The User Interface must follow "Material Design" principles to ensure a consistent look and feel.

NFR-09: The system must provide context-sensitive help (tooltips) for medical coding fields (ICD-10, CPT codes) to assist doctors in billing.

4. Appendix: Data Dictionary (Excerpt)

Data Element	Data Type	Description	Constraints
Patient_ID	String (Alphanumeric)	Unique identifier for the patient.	Primary Key, Auto-Generated.
Prescription_Ref	String (Alphanumeric)	Unique ID for the prescription order.	Must link to a valid Doctor_ID.
Drug_NDC	Integer	National Drug Code for inventory tracking.	10-digit number, mandatory field.
Bill_Total	Decimal (10,2)	Total amount charged to the patient.	Cannot be negative.
Is_Discharged	Boolean	Flags if the IPD patient has been discharged.	Defaults to False.

Document End

This Software Requirements Specification is prepared as a structured requirements baseline for the Cloud-Based Hospital Management System (HMS), Version 2.0.